(CO)COMPLETENESS OF FIBRED DOUBLE CATEGORIES

PhD Writings

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Contents

1 Depend Sums and Products **3**

2 Opfibrations **3**

3 Geometric Morphisms as Fibrations **4**

4 Closure Properties of Fibrations **6**

5 Internal Sums **8**

6 Internal Products **12**

7 Fibred Terminal Objects **13**

8 Fibred Pullbacks **15**

9 Fibrations of 2-categories **16**

10 Fibrant Double Categories **16**

11 Double Fibred Limits **18**
11.1 Double Terminals 18
11.2 Double Fibred Pullbacks 19
11.3 Tabulators and Fibred Tabulators 23

12 Dual Notions **27**
12.1 2-Opfibrations 27
12.2 Double Bifibrations 27
12.2.1 Relation to 2-fibrations 28

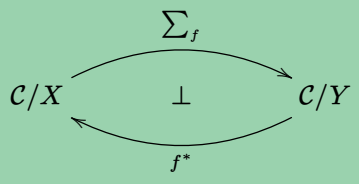
13 Double Internal Products **28**

These are edited notes from [1] including proofs that are originally not found in the text. They cover accordingly Internal Sums and Products.

1 Depend Sums and Products

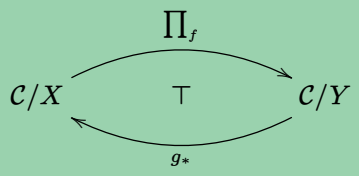
Given a category $\mathcal{C}$ and a morphism morphism $f\colon X \longrightarrow Y$ in $\mathcal{C}$ s.t pullbacks along f exist. We say

Definition 1.1. A depend sum along f is the functor $\sum_f\colon \mathcal{C}/X \longrightarrow \mathcal{C}/Y$ that is left adjoint to the change-of-base functor $f^*\colon \mathcal{C}/Y \longrightarrow \mathcal{C}/X$



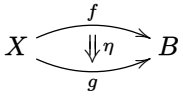
Dually

Definition 1.2. A dependent product along $g\colon X \longrightarrow Y$ is, if it exists, the right adjoint functor $\prod_g\colon \mathcal{C}/X \longrightarrow \mathcal{C}/Y$ to the change-of-base along g.



Remark 1.1. A category with dependent products necessarily has pullbacks. Thus we have

$$\sum_f \dashv f^* \dashv \prod_f$$



2 Opfibrations

Let $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a functor.

1. A morphism $\varphi\colon X \longrightarrow Y$ in $\mathcal{E}$ over $u\colon I \longrightarrow J$ is opcartesian if it is cartesian over u for $P\colon \mathcal{E}^{op} \longrightarrow \mathcal{B}^{op}$. That is, if each morphism $\psi\colon X \longrightarrow Z$ with $P\psi = v \circ u$ in $\mathcal{B}$, uniquely determines a morphism $\theta\colon Y \longrightarrow Z$ over v.

2. A functor $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ is an opfibration if $P\colon \mathcal{E}^{\mathrm{op}} \longrightarrow \mathcal{B}^{\mathrm{op}}$ is a fibration. Equivalently, if for each $u\colon I \longrightarrow J$ in $\mathcal{B}$, there is an opcartesian morphism $\varphi\colon X \longrightarrow Y$ over u.

3. A bifibration is a functor $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ that is a fibration and an opfibration.

Remark 2.1. Usually it is written $u^*X \longrightarrow X$ for a cartesian lifting of $u\colon I \longrightarrow J$. Similarly, it also written $X \longrightarrow u_!X$ or $X \longrightarrow \coprod_u X$ for opcartesian lifting of u. The assignment $X \mapsto u_!X$ extends to a functor which may be called an extension — because it turns out to be left adjoint to u.

Example 2.2. The domain functor $\partial_0 \equiv \mathrm{dom}\colon \mathcal{B}^2 \longrightarrow \mathcal{B}$ is an opfibration if and only if $\mathcal{B}$ has pushouts. Pushouts are precisely the opcartesian morphisms in $\mathcal{B}^2$. The fibre over $I \in \mathcal{B}$ is the opslice $I/\mathcal{B}$.

Lemma 2.3 (B. Jacobs)**.** A fibration is a bifibratoin (i.e. additionally it is an opfibration) if and only if each reindexing functor u^* has a left adjoint, denoted $\coprod_u$.

Proof. Let $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a fibration. For a morphism $u\colon I \longrightarrow J$ in $\mathcal{B}$ and objects $X \in \mathcal{E}_I$ and $Y \in \mathcal{E}_J$ consider the following chain

$$\mathcal{E}_J\left(\coprod_u X, Y\right) \overset{*}{\cong} \mathcal{E}_I(X, u^*Y) \cong \mathcal{E}_u(X, Y) \overset{**}{\cong} \mathcal{E}_J\left(\coprod_u X, Y\right)$$

where $\cong$ is due to P being a fibration. Then

$$\begin{aligned} \text{left adjoints } \coprod_u \text{ exist} &\iff \overset{*}{\cong} \text{ exist} \\ &\iff \overset{**}{\cong} \text{ exist} \\ &\iff P \text{ is an opfibrati on.} \end{aligned}$$

□

Remark 2.4. The Beck-Chevalley condition is not required for these $\coprod_u$'s in the above lemma. However, this result implies that every fibration with sums (i.e. adjunctions $\coprod_u \dashv u^*$ satisying Beck-Chevalley) is a bifibration. Thus, codomain fibrations are bifibrations.

3 Geometric Morphisms as Fibrations

Theorem 3.1. Given a functor $F\colon \mathcal{C} \longrightarrow \mathcal{D}$ with a fully faithful right adjoint G s.t F preserves finite limits and $\mathcal{C}$ is finitely complete, then F is a Street fibration.

Before we prove the above theorem, we provide this lemma

Lemma 3.2. If a functor $G\colon \mathcal{D} \longrightarrow \mathcal{C}$ has a left adjoint F with counit ε. Then G is fully faithful iff the ε is an isomorphism.

Proof. Consider

$$\begin{array}{ccc} \mathrm{Hom}(X, X') & \xrightarrow{\ G\ } & \mathrm{Hom}(GX, GX') \\ & \searrow^{\varepsilon_X} & \Big\downarrow \sim \ \mathrm{ad} \\ & & \mathrm{Hom}(FGX, X') \end{array}$$

then $\mathrm{Hom}(X, X') \xrightarrow{\sim} \mathrm{Hom}(GX, GX')$ iff ε_X is a bijection. G is then fully faithful iff $FGX \longrightarrow X$ is an isomorphism for all X. □

We can now prove the theorem

Proof. Given Y in $\mathcal{C}$ with $F(Y) = J$ and a morphism $u\colon I \to J$, let u^*Y be the pullback of the unit $\eta\colon Y \longrightarrow G(J) = GF(Y)$ along $G(u)\colon G(I) \longrightarrow G(J)$. Since $F(\eta_Z)$ is an isomorphism (as G is fully faithful) and F preserves pullbacks, F also sends the morphism $u^*(Y) \longrightarrow G(I)$ to an isomorphism. As illustrated in the diagram below

$$\begin{array}{ccc} u^*Y & \xrightarrow{\ \varphi\ } & Y \\ \Big\downarrow & & \Big\downarrow{\scriptstyle \eta_Y} \\ G(I) & \xrightarrow[G(u)]{} & G(J) \end{array}$$

Claim: $\varphi\colon u^*Y \longrightarrow Y$ is cartesian.
Suppose $\psi\colon Z \longrightarrow Y$ is a morphism over $v\colon K \longrightarrow I$ and ψ is over $u \circ v$. Then we have $Gu \circ Gv \circ \eta_Z = \eta_Y \circ \psi$ by the naturality of the adjunction unit. The universal property of the pullback gives a unique morphism $\theta\colon Z \longrightarrow u^*Y$ over v. As shown below

$$\begin{array}{ccccc} & & \overset{\psi}{\frown} & & \\ Z & \overset{\theta}{\dashrightarrow} & u^*Y & \xrightarrow{\ \phi\ } & Y \\ \Big\downarrow{\scriptstyle \eta_Z} & & \Big\downarrow & & \Big\downarrow{\scriptstyle \eta_Y} \\ G(K) & \xrightarrow[G(v)]{} & G(I) & \xrightarrow[G(u)]{} & G(J) \end{array}$$

$$K \xrightarrow{\ v\ } I \xrightarrow{\ u\ } J$$

Furthermore, for any other morphism $\tilde{\theta}\colon Z \longrightarrow u^*Y$ over v that commutes with ψ and φ, it must satisfy $G(v) \circ \eta_Z = \eta \circ \tilde{\theta}$, by the naturality of the adjunction it must equal θ. □

4 Closure Properties of Fibrations

Definition 4.1. Let $P\colon X \longrightarrow Y$ be a functor. A morphism $f\colon x \longrightarrow y$ in X is cartesian if given a morphism $h\colon z \longrightarrow y$ with $Ph = Pf \circ u$ where $u\colon v \longrightarrow Px$ there exists a unique morphism $g\colon z \longrightarrow x$ s.t $Pg = u$ and $f \circ g = h$.

Definition 4.2. A functor $P\colon X \longrightarrow Y$ is a fibration if for each $x \in X$ and a morphism $f\colon y \longrightarrow Px$ in Y there is a cartesian morphism $f'\colon f^*x \longrightarrow x$ s.t $Pf' \cong f$.

Theorem 4.1. Let $P\colon X \to B$ be a fibration and $F\colon Y \longrightarrow X$ be an arbitrary functor. Then F itself is a fibration iff

1. $Q \cong P \circ F$ is a fibration and F is a cartesian functor from Q to P and
2. Each $F_I\colon Y_I \longrightarrow X_I$ is a fibration and cartesian morphisms w.r.t these fibrations are stable under reindexing i.e., for every commutative square

$$\begin{array}{ccc} Y_1 & \xrightarrow{\varphi_1} & X_1 \\ {\scriptstyle\theta}\downarrow & & \downarrow{\scriptstyle\psi} \\ Y_2 & \xrightarrow[\varphi_2]{} & Y_1 \end{array}$$

in Y with φ_1 and φ_2 cartesian w.r.t Q over $u\colon J \longrightarrow I$ and $Q(\theta) = id_J$ and $Q(\psi) = id_I$ it holds that θ is cartesian w.r.t F_J whenever ψ is cartesian w.r.t F_I.

Proof. ($\Longrightarrow$) Since fibration are closed under composition it follows that $Q \equiv P \circ F$ is a fibration. Consider a $Q-$cartesian morphism $\varphi\colon z \longrightarrow y$ over $P(f)$ where $f\colon x \longrightarrow Fy$ is $P-$cartesian. Then $PF(\varphi) \cong Q(\varphi) \cong P(f)$, we have two morphisms $F(\varphi)$ and f both to $F(\varphi)$ over $P(f)$, it follows that $F(\varphi) \cong f$. To show 2, let $y \in Y_I$ and $u\colon x \longrightarrow Fy$ in X_I. There is a $F-$cartesian morphism $w\colon e \longrightarrow e'$ in Y with $F(w) \cong u$. It remains to show that $w \in Y_I$. Since $u \in X_I$ we have $P(w) \cong id_I$ and $PF(w) \cong P(u) \cong id \implies w \in Y_I$.
($\Longleftarrow$) Let $y \in Y_I$ lying over $x \in X$ ($x \cong F(y)$) and both x and y lie over $b \in B$ ($b \cong P(x) \cong Q(y)$) and morphisms $\delta\colon x' \longrightarrow x$ in X and $\beta\colon b' \longrightarrow b$ in B. We are required to show: there is a cartesian morphism $y' \longrightarrow y$ lying over δ. But since Q is a fibration, there is a $Q-$cartesian morphism $\varphi\colon \beta^*y \longrightarrow y$ in Y and $F-$cartesian implies that $F(\varphi)\colon \beta^*x \longrightarrow x$ is a $P-$cartesian morphism.
Notice that there is a unique factorization of δ through $F(\varphi)$

$$x' \xrightarrow{\alpha} \beta^*x \xrightarrow{F(\varphi)} x$$

and α is a morphism in b'. Since $F_{b'}$ is a fibration, there is a $F_{b'}-$cartesian morphism $\psi\colon \alpha^*\beta^*y \longrightarrow \beta^*y$ in $Y_{b'}$. We must show that the composite

$$\varphi \circ \psi\colon \alpha^*\beta^*y \longrightarrow \beta^*y \longrightarrow y$$

is cartesian over δ. Consider the following diagram

$$\begin{array}{ccccc}
Maps_Y(y', \alpha^*\beta^* y) & \longrightarrow & Maps_Y(y', \beta^* y) & \longrightarrow & Maps_Y(y', y) \\
\downarrow & (1) & \downarrow & (2) & \downarrow \\
Maps_X(F(y'), x') & \longrightarrow & Maps_X(F(y'), \beta^* x) & \longrightarrow & Maps_X(F(y'), x) \\
\downarrow & (3) & \downarrow & (4) & \downarrow \\
Maps_B(Q(y'), b') & \xrightarrow[\text{id}]{} & Maps_B(Q(y'), b') & \longrightarrow & Maps_B(Q(y'), b)
\end{array}$$

we must show that the composite of the two upper squares is cartesian. To do this, we shall show that both (1) and (2) are cartesian. By our construction φ is $Q-$cartesian and $F\varphi$) is $P-$cartesian, so the composite of (2) and (4) is cartesian and (4) is cartesian, so (2) must be cartesian. To prove that (1) is cartesian, we know that a commutative square is cartesian iff the induced maps on the fibers are equivalences, it suffices to show that the square

$$\begin{array}{ccc}
Maps_Y(y', \alpha^*\beta^* y)_k & \longrightarrow & Maps_Y(y', \beta^* y)_k \\
\downarrow & & \downarrow \\
Maps_X(F(y'), x')_k & \longrightarrow & Maps_X(F(y'), \beta^* x)_k
\end{array}$$

obtained by taking the fiber at $k\colon Q(y') \longrightarrow b'$ is cartesian for all maps k. Taking $P-$ and $Q-$cartesian pullbacks along k we can (since F is cartesian functor) identify such a square with the following square

$$\begin{array}{ccc}
Maps_{Y_{Q(y')}}(y', k^*\alpha^*\beta^* y) & \longrightarrow & Maps_{Y_{Q(y')}}(y', k^*\beta^* y) \\
\downarrow & & \downarrow \\
Maps_{X_{Q(y')}}(F(y'), k^* x') & \longrightarrow & Maps_{X_{Q(y')}}(F(y'), k^*\beta^* x)
\end{array}$$

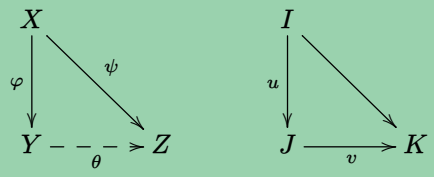
But this is cartesian since by our assumption the map

$$k^*\alpha^*\beta^* y \longrightarrow k^*\beta^* y$$

is $F_{Q(y')}-$cartesian (because $\psi\colon \alpha^*\beta^* y \longrightarrow \beta^* y$ is $F_{b'}-$cartesian). $\square$

5 Internal Sums

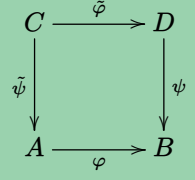
Definition 5.1. Let $\mathcal{B}$ be a category with pullbacks and $P\colon \mathcal{X} \longrightarrow \mathcal{B}$ a fibration. A morphism $\varphi\colon X \longrightarrow Y$ over $u\colon I \longrightarrow J$ is called cocartesian iff for every $v\colon J \longrightarrow K$ in $\mathcal{B}$ and $\psi\colon X \longrightarrow Z$ over $v \circ u$ there is a unique morphism $\theta\colon Y \longrightarrow Z$ over v with $\psi = \theta \circ \varphi$.



The fibration P has internal sums iff the following conditions are satisfied.

1. For every $X \in P(I)$ and $u\colon I \longrightarrow J$ in $\mathcal{B}$ there is a cocartesian morphism $\varphi\colon X \longrightarrow Y$ over u.

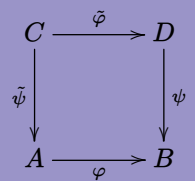
2. The Beck-Chevalley Condition (BCC) holds i.e., for every commuting square in $\mathcal{X}$



over a pullback in $\mathcal{B}$, it holds that $\tilde{\varphi}$ is cocartesian whenever φ is cocartesian and ψ and $\tilde{\psi}$ are cartesian.

Remark 5.1. 1. It is easily seen that for a fibration $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ a morphism $\varphi\colon X \longrightarrow Y$ is cocartesian iff for all $\psi\colon X \to Z$ over $P(\varphi)$ there exists a unique vertical morphism $\alpha\colon Y \to Z$ with $\alpha \circ \varphi = \psi$.

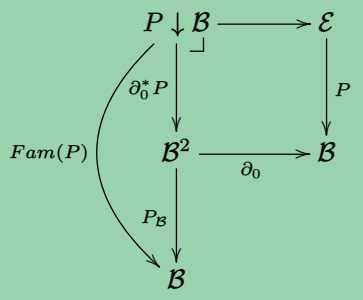
2. One easily sees that the BCC of Definition 5.1 is equivalent to the requirement that for every commuting square in $\mathcal{E}$



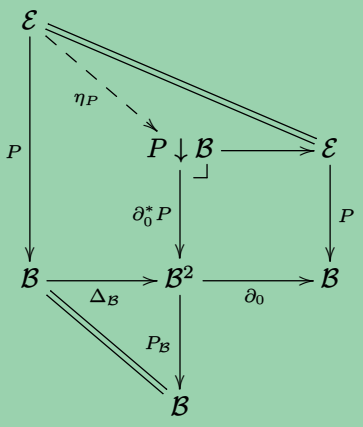
over a pullback in $\mathcal{B}$ it holds that ψ is cartesian whenever $\tilde{\psi}$ is and φ and $\tilde{\varphi}$ are cocartesian.

We now give an explanation for internal sums in which the Beck-Chevalley Condition arises in a less special way. To this end, we generalize the Fam construction from ordinary categories to fibrations.

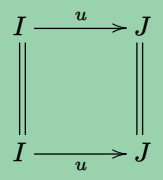
Definition 5.2. Let $\mathcal{B}$ be a category with pullbacks and $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a fibration. Then the family fibration $Fam(P)$ for P is defined as $P_{\mathcal{B}} \circ \partial_0^* P$ where



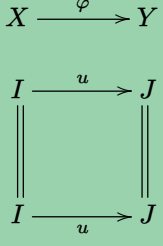
is a pullback. The cartesian functor $Fam(P)\colon \partial_0^* P \longrightarrow P_{\mathcal{B}}$ is called the fibred family fibration for P.
We can define the cartesian functor $\eta_P\colon P \longrightarrow Fam(P)$ as in the diagram



where $\Delta_{\mathcal{B}}$ sends a morphism $u\colon I \longrightarrow J$ in $\mathcal{E}$ to

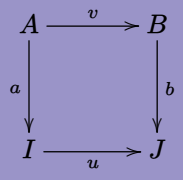


in $\mathcal{B}^2$. More explicitly, η_P sends $\varphi\colon X \longrightarrow Y$ over $u\colon I \longrightarrow J$ to a pair

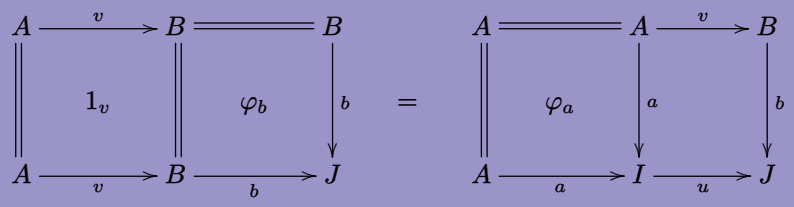


in $P \downarrow \mathcal{B}$. Clearly the functor η_P preserves the cartesianness of morphisms i.e., η_P is cartesian.

Remark 5.2. If a morphism $Fam(P)(\varphi)$ is $P_{\mathcal{B}}$-cocartesian the φ is $Fam(P)$-cartesian iff φ is $Fam(P)$-cocartesian. Moreover, for every morphism



in $\mathcal{B}^2$ we have

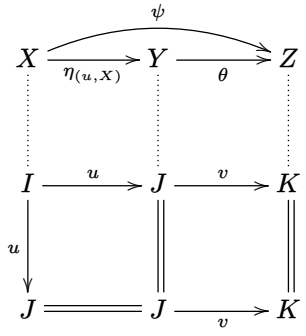


where φ_a and φ_b are $P_{\mathcal{B}}$-cocartesian. From these two observations, one can show that for fibrations P and Q over $\mathcal{B}$ a cartesian functor $F\colon Fam(P) \longrightarrow Fam(Q)$ is determined uniquely up to isomorphism by its restriction along the inclusion functor $\Delta_{\mathcal{B}}$ from which it follows that $F \cong Fam(\Delta_B{}^* F)$. Thus up to isomorphism all cartesian functors from $Fam(P)$ to $Fam(Q)$ are of the form $Fam(F)$ for some cartesian functor $F\colon P \longrightarrow Q$.

Next we characterize the property of having internal sums in terms of the family monad Fam. [1].

Theorem 5.3. Let $\mathcal{B}$ be a category with pullbacks and $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a fibration. Then P has internal sums iff $\eta_P\colon P \longrightarrow Fam(P)$ has a fibred left adjoint $\coprod_P\colon Fam(P) \longrightarrow P$ i.e., $\coprod \dashv \eta_P$ where $\coprod_P$ is cartesian and unit and counit of the adjunction are cartesian natural transformations.

Proof. ($\Longleftarrow$) We explicitate the universal property of the unit of the adjunction $\coprod_P \dashv \eta_P$ at an object (u, X) in the following diagram



[1] One can show that η and μ are natural transformations giving rise to a monad (Fam, η, μ) on **Fib**($\mathcal{B}$)

whose left column is the unit of the adjunction at (u, X). From this it follows that $\eta_{(u,X)} \colon X \longrightarrow Y$ is cocartesian over u. Cartesianness of $\coprod_P$ says that the cartesian arrow f as given by

$$\begin{array}{ccc} X & \xrightarrow{\psi} & Y \\ & \text{cart} & \\ K & \xrightarrow{q} & L \\ {\scriptstyle p}\downarrow & \lrcorner & \downarrow{\scriptstyle v} \\ I & \xrightarrow[u]{} & J \end{array}$$

in $P \downarrow \mathcal{B}$ is sent by $\coprod_P$ to the cartesian arrow $\coprod_P f$ over u satisfying

$$\begin{array}{ccc} X & \xrightarrow{\psi} & Y \\ & \text{cart} & \\ {\scriptstyle \eta_{(u,X)}}\downarrow & & \downarrow{\scriptstyle \eta_{(v,Y)}} \\ & \text{cart} & \\ A & \xrightarrow[\coprod_P f]{} & B \end{array}$$

where $\eta_{(u,X)}$ and $\eta_{(v,Y)}$ are cocartesian units of the adjunction above p and v respectively. Thus, by (2) of Remark 5.1 cartesianness of $\coprod_P$ is just the Beck-Chevalley Condition for internal sums.

($\Longrightarrow$) If P has internal sums then the functor $\coprod_P$ left adjoint to P is given by sending a morphism f in $P \downarrow \mathcal{B}$ as given by

$$\begin{array}{ccc} X & \xrightarrow{\psi} & Y \\ & & \\ K & \xrightarrow{q} & L \\ {\scriptstyle p}\downarrow & & \downarrow{\scriptstyle v} \\ I & \xrightarrow[u]{} & J \end{array}$$

to the morphism $\coprod_P f$ over u satisfying

$$\begin{array}{ccc} X & \xrightarrow{\psi} & Y \\ {\scriptstyle \varphi_1}\downarrow & & \downarrow{\scriptstyle \varphi_2} \\ A & \xrightarrow[\coprod_P f]{} & B \end{array}$$

where φ_1 and φ_2 are cocartesian over p and v respectively. One can verify that indeed $\coprod_P$ is left adjoint to η_P using the units of the adjunction the cocartesian liftings guaranteed for P. Again, cartesianness of $\coprod_P$ is easily seen to be equivalent to the Beck-Chevalley Condition.

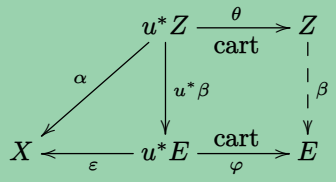
□

The next chapter discusses the dual of internal sums.

6 Internal Products

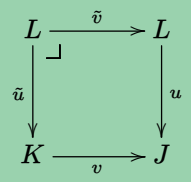
Definition 6.1. Let $\mathcal{B}$ be a category with pullbacks. A fibration $P\colon \mathcal{X} \longrightarrow \mathcal{B}$ has internal products if the following hold:

1. For $u\colon I \longrightarrow J$ in $\mathcal{B}$ and $X \in P(I)$



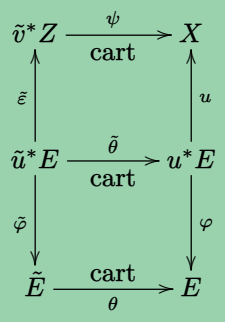
   The span (φ, ε) is the determined uniquely up to vertical isomorphism. It is called an evaluation span for X along u.

2. Whenever the pullback



   in $\mathcal{B}$ and (φ, ε) is an evaluation span for X along u then

   with the lower square above the pullback in $\mathcal{B}$, $\tilde{\varepsilon}$ vertical, then $(\tilde{\varphi}, \tilde{\varepsilon})$ is an evaluation span for $\tilde{v}^*X$ along v.



Remark 6.1. Condition (2) is called the Beck-Chevalley Condition for internal products and essentially says that evaluation spans are stable under reindexing.

We now give a characterization of internal sums and products in terms of adjunctions from §1 in [2].

Lemma 6.2. Let $\mathcal{B}$ be a category with pullbacks and $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a fibration. We say P has internal products (resp. sums) iff the following are satisfied

1. for every morphism $u\colon I \longrightarrow J$ in $\mathcal{B}$, the change of base functor $u^*\colon \mathcal{E}_J \longrightarrow \mathcal{E}_I$ has a right adjoint (resp. left adjoint).

2. The Beck-Chevalley Condition holds: For every pullback in $\mathcal{B}$ of the form

$$\begin{array}{ccc} K & \xrightarrow{v} & L \\ {\scriptstyle r}\downarrow & \lrcorner & \downarrow{\scriptstyle s} \\ I & \xrightarrow[u]{} & J \end{array}$$

the canonical natural transformation

$$s^* \prod_u \Longrightarrow \prod_v r^* \qquad \left(\text{resp. } \coprod_v r^* \Longrightarrow s^* \coprod_u\right)$$

is an isomorphism.

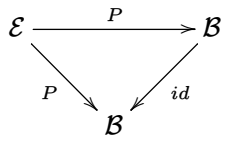
7 Fibred Terminal Objects

Definition 7.1. A fibration $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ is a fibration of categories with terminal object if and only if every fibre $P(I)$ has a terminal object and these are stable reindexing.

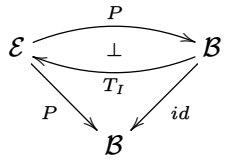
Lemma 7.1. Let $\mathcal{B}$ have a terminal object (besides having pullbacks). A fibration $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ is a fibration of categories with terminal objects if and only if $\mathcal{E}$ has a terminal $1_\mathcal{E}$ and $P(1_\mathcal{E})$ is terminal in $\mathcal{B}$.

Proof. Suppose $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ is a fibration of categories with terminal objects. Let $1_\mathcal{E}$ be a fibred terminal over $1_\mathcal{B}$ and $u\colon I \longrightarrow 1_\mathcal{B}$. Let $\bar{u}(1_\mathcal{E})\colon u^*1_\mathcal{E} \longrightarrow 1_\mathcal{E}$ be the cartesian lift of u. Then $u^*1_\mathcal{E}$ is a terminal in $P(I)$ and for any $i \in P(I)$ there is a unique vertical arrow $h\colon i \longrightarrow u^*1_\mathcal{E}$ with $\bar{u} \circ h$ over u. We must show that $1_\mathcal{E}$ is terminal in $\mathcal{E}$. Suppose $f\colon i \longrightarrow 1_\mathcal{E}$ is another morphism. Then by the universal property of cartesian morphisms, $f = \bar{u} \circ h$. Thus $\bar{u} \circ h$ is the unique arrow from i to $1_\mathcal{E}$. Conversely, suppose $1_\mathcal{E}$ is terminal in $\mathcal{E}$ with $P(1_\mathcal{E})$ terminal in $\mathcal{B}$. Let $u\colon I \longrightarrow 1_\mathcal{B}$ the unique morphism in $\mathcal{B}$ and $\bar{u}(1_\mathcal{E})\colon u^*1_\mathcal{E} \longrightarrow 1_\mathcal{E}$ be the cartesian lift of u. We must show that $u^*1_\mathcal{E}$ is terminal in $P(I)$. To this end, let $a \in P(I)$ and $g\colon a \longrightarrow 1_\mathcal{E}$ be the unique morphism in $\mathcal{E}$. Since $\bar{u}$ is cartesian, there is a unique vertical morphism $l\colon a \longrightarrow u^*1_\mathcal{E}$ such that $\bar{u} \circ l = g$. For any l' with $\bar{u} \circ l' = g$ it follows that $l = l'$ since $1_\mathcal{E}$ is terminal. This shows that $u^*1_\mathcal{E}$ is terminal in $P(I)$. □

Let $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a fibration. In the category $\mathbf{Fib}/\mathcal{B}$, P has a terminal object if P

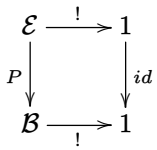


has a right-adjoint. As illustrated below.

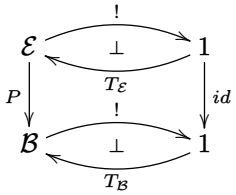


This fibrewise terminal T_I is preserved by cartesianness.

In **Fib**, P has a terminal object if



has a right-adjoint in **Fib**

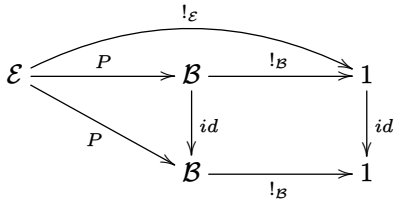


i.e., both $\mathcal{E}$ and $\mathcal{B}$ have terminal objects and P preserves them.

Theorem 7.2. If $\mathcal{B}$ has a terminal object, then the following are equivalent

1. P has a terminal object in $\mathbf{Fib}/B$
2. P has a terminal object in **Fib**.

Proof. By the lifting-and-factorization of adjoints



Given that $!_{\mathcal{B}} \dashv T_{\mathcal{B}}$

i. P has a right-adjoint, $P \dashv T_{\mathcal{E}}$,

ii. $!_{\mathcal{E}}$ has a right-adjoint $!_{\mathcal{E}} \dashv T_{\mathcal{E}}$ over $!_{\mathcal{B}} \dashv T_{\mathcal{B}}$

□

8 Fibred Pullbacks

Let $\mathcal{B}$ be a category with pullbacks.

Lemma 8.1. For a fibration $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ we have

1. A commuting square of cartesian arrows is always a pullback in $\mathcal{E}$
2. A commuting square

$$\begin{array}{ccc} Y_1 & \xrightarrow[\text{cart}]{\varphi_1} & X_1 \\ {\scriptstyle\beta}\downarrow & & \downarrow{\scriptstyle\alpha} \\ Y_2 & \xrightarrow[\varphi_2]{\text{cart}} & X_2 \end{array}$$

in $\mathcal{E}$ with α and β vertical and φ_i cartesian is always a pullback.

Proof. (2) Consider a cone

$$\begin{array}{ccc} Q & \longrightarrow & X_1 \\ \downarrow & & \downarrow \\ Y_2 & \longrightarrow & X_2 \end{array}$$

over the cospan $Y_2 \longrightarrow X_2 \longleftarrow X_1$, then its image is

$$\begin{array}{ccc} P(Q) & & \\ \downarrow & \searrow & \\ P(Y_2) & \xrightarrow[P(\varphi_2)]{} & P(X_2) \end{array}$$

since $P(\varphi_1) = P(\varphi_2)$, another lift is given by

$$\begin{array}{ccc} Q & & \\ \vdots & \searrow & \\ Y_1 & \xrightarrow[\varphi_1]{} & X_1 \end{array}$$

then there exists a unique $Q \longrightarrow Y_1$ since φ_1 is cartesian. But we have two *cartesian fillers*, namely, $Q \longrightarrow Y_2$ and the $Q \longrightarrow Y_1 \longrightarrow Y_2$ over

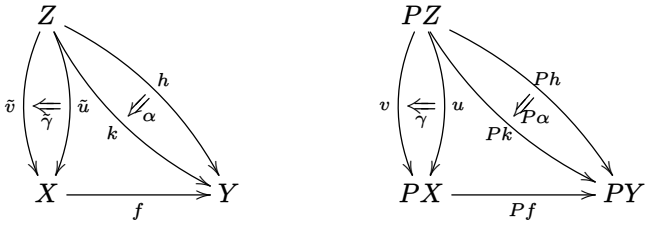
$$\begin{array}{ccc} P(Q) & & \\ \downarrow & \searrow & \\ P(Y_2) & \xrightarrow[P(\varphi_2)]{} & P(X_2) \end{array}$$

since φ_2 is cartesian, these unique morphisms must be equal. Thus $Q \longrightarrow Y_1$ is a cone morphism and unique. Hence the square is a pullback. □

9 Fibrations of 2-categories

We recall some basic aspects of fibred 2-categories from [3]

Definition 9.1. Let $P\colon \mathcal{E} \longrightarrow \mathcal{B}$ be a 2-functor. A morphism $f\colon X \longrightarrow Y$ is said to be cartesian if it is cartesian in a sense of Definition 4.1 and additionally, for any 2-cells $\sigma\colon h \Longrightarrow k\colon Z \longrightarrow Y$ and $\gamma\colon v \Longrightarrow u\colon PZ \longrightarrow PX$, and the lifts $\tilde{v}$ and $\tilde{u}$ of v and u exist, then there is a unique 2-cell $\tilde{\gamma}\colon \tilde{v} \Longrightarrow \tilde{u}\colon Z \longrightarrow X$.



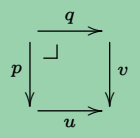
10 Fibrant Double Categories

A double category $\mathbb{D}$ is said to have vertical pullbacks if the structure

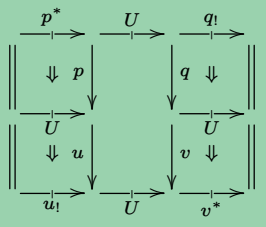
$$\mathcal{D}_1 \times_{\mathcal{D}_0} \mathcal{D}_1 \longrightarrow\!\!\!\!\!\odot\!\!\!\!\!\rightarrow \mathcal{D}_1 \underset{t}{\overset{s}{\underset{\longleftarrow i \longrightarrow}{\rightrightarrows}}} \mathcal{D}_0$$

lies in **PBCat**, the category of categories with pullbacks and pullback-preserving functors. Given a pullback in $\mathcal{D}_0$ we can get the corresponding square of unit of cells in $\mathcal{D}_1$ which is a pullback. $\odot$ preserves pullbacks i.e., we can horizontally compose two vertical pullback squares and obtain a vertical pullback sqaure. This is essentially having a functorial choice of pullbacks.

Definition 10.1 (Beck-Chevalley Condition)**.** For each pullback of vertical arrows



the cell

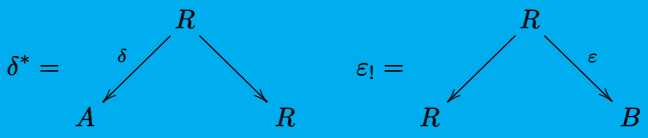


is invertible.

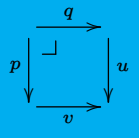
Definition 10.2. [4] A fibrant double category $\mathbb{D}$ has internal sums whenever the fibration $(s,t)\colon \mathcal{D}_1 \longrightarrow \mathcal{D}_0 \times \mathcal{D}_0$ has internal sums i.e., opcartesian lifts of (s,t) satisfy the Beck-Chevalley Condition.

Example 10.1. [5] $\mathbb{S}\mathrm{pan}(\mathcal{C})$ as a fibrant double category has internal sums. For a vertical arrow $f\colon X \longrightarrow Y$, the extension and restriction along f are given by $X \xleftarrow{id} X \xrightarrow{f} Y$ and $Y \xleftarrow{f} X \xrightarrow{id} X$ respectively.

Note that any span $R\colon A \nrightarrow B$ can be factors as $\delta^* \circ \varepsilon_!$ where

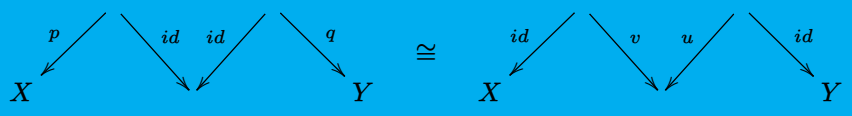


More generally we have two embeddings $_!\colon \mathcal{C} \longrightarrow \mathbb{S}\mathrm{pan}(\mathcal{C})$ and $^*\colon \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbb{S}\mathrm{pan}(\mathcal{C})$ such that $f_! \dashv f^*$ for $f\colon X \longrightarrow Y$. These right adjoints satisfy the Beck-Chevalley Condition



$$\Longrightarrow \quad q_! p^* \cong u^* v_!$$

i.e.



Remark 10.2. The above span factorization property is inherited by $\mathbb{R}\mathrm{el}$ as a bicategory. Similarly, we can express a relation $R\colon\ X \nrightarrow Y$ from X to Y as a span. Composition is given by pullback and 2-cells are just the morphism between the two top objects of the spans. Whence both $\mathbb{R}\mathrm{el}$ and $\mathbb{S}\mathbf{pan}(\mathcal{C})$ have internal sums. The Beck-Chevalley Cell in Definition 10.1 can be seen from the example above.

11 Double Fibred Limits

Fix $\mathbb{B}$ a pseudo category in **PBCat**, a category of categories with pullbacks and pullback-preserving functors.

11.1 Double Terminals

Consider a double category $\mathbb{D}$ as a pseudo category in **TCat**, the category of categories with terminal objects. A terminal object T in $\mathbb{D}_0$ gives rise to a terminal object U_T in $\mathbb{D}_1$ such that for any proarrow $A \overset{M}{\nrightarrow} B$ there is precisely one cell

$$\begin{array}{ccc} A & \overset{M}{\nrightarrow} & B \\ \downarrow & \theta & \downarrow \\ T & \underset{U_T}{\nrightarrow} & T \end{array}$$

from M to U_T, for any object $M \in \mathbb{D}_1$. This is referred to as the 2-dimensional property of a terminal object.

Definition 11.1. A strict double functor $\mathcal{F}\colon \mathbb{E} \longrightarrow \mathbb{B}$ between pseudo double categories is a double fibration of double categories with terminal objects iff every fibre has a terminal object for $\mathcal{F}_0$ and $\mathcal{F}_1$.

Remark 11.1. $\mathcal{F}_0\colon \mathbb{E}_0 \longrightarrow \mathbb{B}_0$ is an ordinary fibration of categories with terminal objects.

Let us explicate the fibre at level 1: Let $M \in \mathbb{B}_1$ be an object, the fibre $\mathcal{F}_1(M)$ consists of proarrows N such that $\mathcal{F}_1 N = M$ and cells

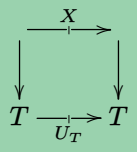
$$\begin{array}{ccc} & \overset{N}{\nrightarrow} & \\ \downarrow & \theta & \downarrow \\ & \underset{L}{\nrightarrow} & \end{array}$$

over the globular cell

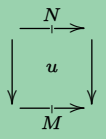
$$\begin{array}{ccc} & \overset{M}{\nrightarrow} & \\ \| & \mathcal{F}_1\theta & \| \\ & \underset{M}{\nrightarrow} & \end{array}$$

Remark 11.2. We shall call cells in the fibre *vertical cells.*

Definition 11.2. $\mathcal{F}_1 \colon \mathbb{E}_1 \longrightarrow \mathbb{B}_1$ is a fibration of categories with terminal objects in the sense that a terminal in the fibre of M is an object $T \xrightarrow{U_T} T$ such that for any other object X in the fibre is a unique cell



over



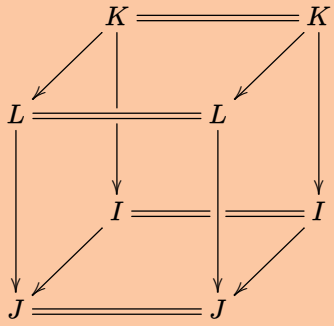
Proposition 11.3. Let $\mathbb{B}$ also have terminal objects. A strict double fibration $\mathcal{F} \colon \mathbb{E} \longrightarrow \mathbb{B}$ is a double fibration of double categories with terminal objects if and only if $\mathbb{E}$ has a terminal T with FT terminal in $\mathbb{B}$.

Proof. Straightfoward application of Lemma (7.1) in 2-dimensional way.

□

11.2 Double Fibred Pullbacks

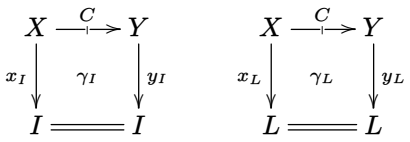
Proposition 11.4. Consider a commuting cube



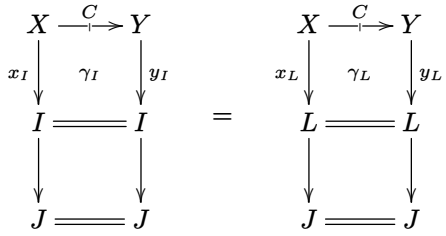
where the top and bottom faces are cartesian cells and the front and back faces are vertical is always a pullback in $\mathbb{E}_1$. We shall write $(\mathbf{T}), (\mathbf{F}), (\mathbf{B}), (\mathbf{L}), (\mathbf{R}), (\mathbf{G})$ for the top, front, back, left, right, and bottom faces respectively.

Proof. The $(\mathbf{T})$ and $(\mathbf{G})$ faces are cartesian since the identity functor $i \colon \mathbb{E}_0 \longrightarrow \mathbb{E}_1$ pre-

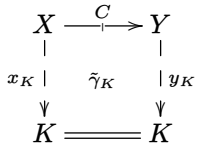
serves cartesian morphisms. For a double cone C over (**F**) and (**G**) we have



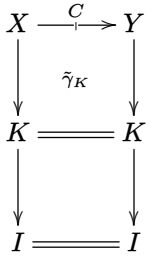
such that



since (**T**) is cartesian there is a unique cell



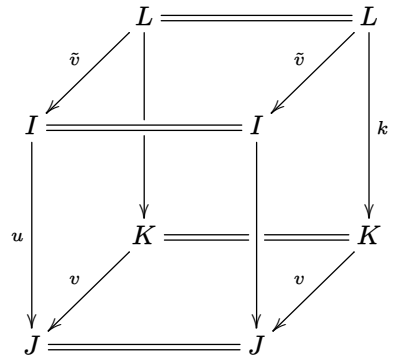
there is another unique cell $\tilde{\gamma}_I$ that we started with given by the composition



Since (**G**) is cartesian, these unique cells must be equal. Thus the cell $\tilde{\gamma}_K$ is the unique cone morphism. □

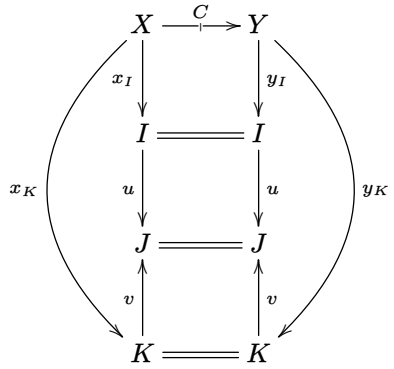
Proposition 11.5. If $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$ is a double fibration of double categories with pullbacks then every pullback in a fibre is a pullback in $\mathbb{E}$.

Proof. Clearly, this is true for the fibration $\mathcal{P}_0$. To show this for $\mathcal{P}_1$, let

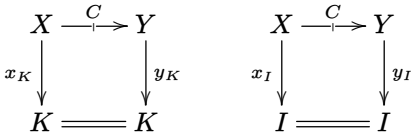


(†)

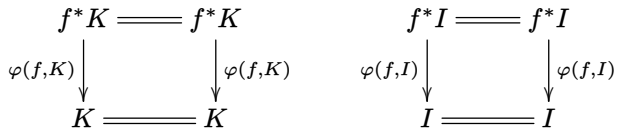
be a pullback in $\mathbb{E}_1$ over $\xrightarrow{M} \in \mathbb{B}_1$. Consider the following diagram



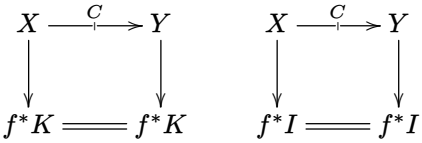
with $v \circ x_K = u \circ x_I$ and $v \circ y_K = u \circ y_I$ for an object $C \in \mathbb{E}_1$ i.e., a cone over (**F**) and (**G**). The cells



lie over the same cell $\theta \colon N \longrightarrow M$ in $\mathbb{B}_1$. Let the cells

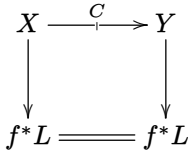


be cartesian i.e., the vertical arrows are cartesian lifts of f in the base. Moreover, let

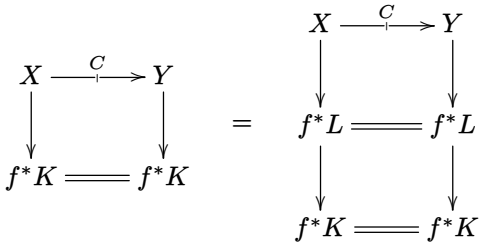


be vertical cells over $\xrightarrow{N}$. We know that (†) is a pullback under change of base, there

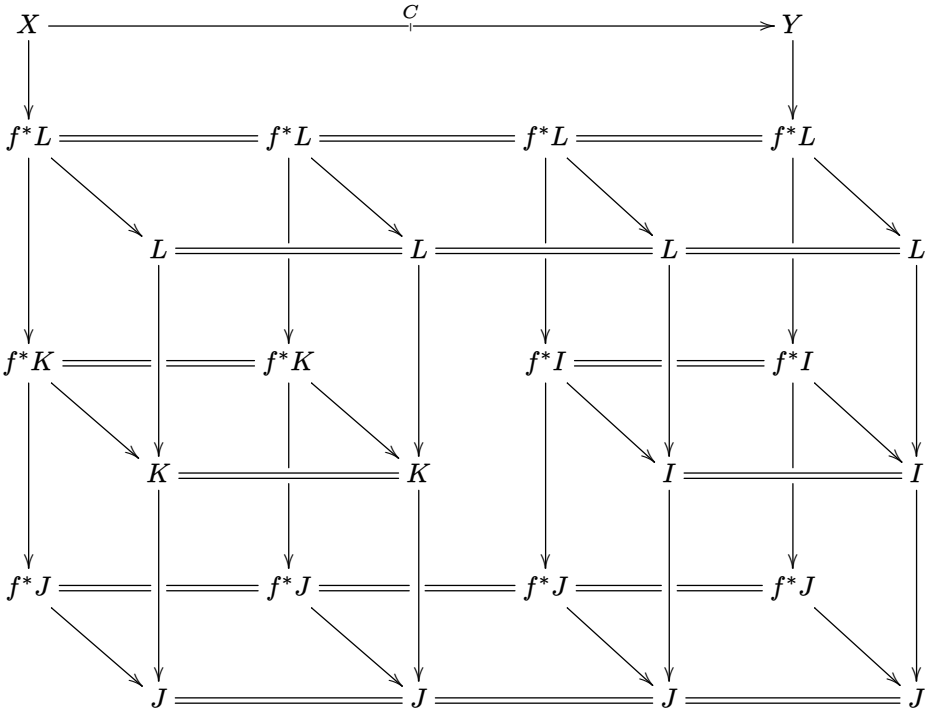
is a vertical cell



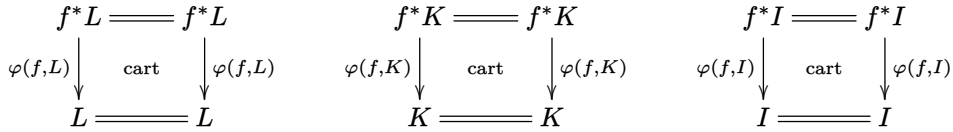
with



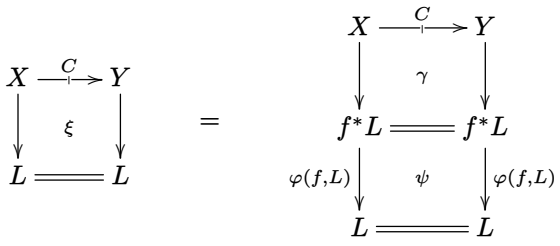
similarly for f^*I. We obtain the following diagram



where the cells



are cartesian. The desired cell is given by

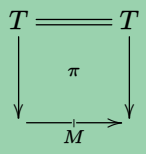


For any other cell $\xi' = \frac{\gamma'}{\psi}$. It follows that $\gamma' = \gamma$ as both are mediating arrows to the pullback under change of base for the cone given by vertical cells into f^*K and f^*I and therefore, it follows that $\xi = \xi'$.

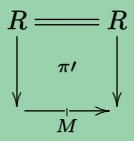
□

11.3 Tabulators and Fibred Tabulators

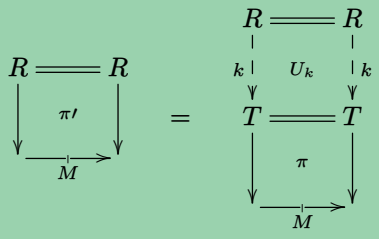
Definition 11.3. In a double category $\mathbb{D}$, a *1-tabulator* of a proarrow M is an object $1^{\bullet}_{T}$ together with a cell



such that for any other cell



we have

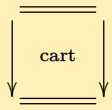


Lemma 11.6 (Grandis-Parè)**.** A double category $\mathbb{D}$ has 1-tabulators if and only if the functor $i\colon \mathbb{D}_0 \longrightarrow \mathbb{D}_1$ has a right adjoint.

We shall refer to 1-tabulators with tabulators only.

Definition 11.4. Let $\mathbb{B}$ be a double category with tabulators. A strict double fibration $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$ is double fibration of double categories with tabulators if and only if every fibre has tabulators and they are stable reindexing.

Lemma 11.7. For a double fibration $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$, a cartesian cells of the form

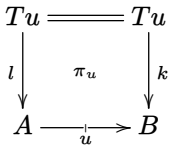


can be reduced to tabulators of proarrow identities in $\mathbb{E}$.

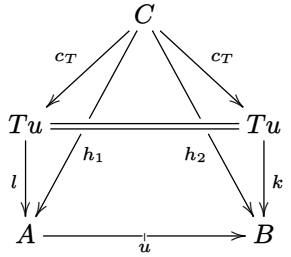
Proof. Immediate from Definition 11.3. □

Lemma 11.8. For a double fibration $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$ of double categories with tabulators, a tabulator in a fibre is a tabulator in $\mathbb{E}$.

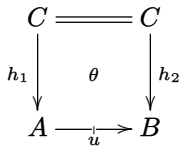
Proof. Let



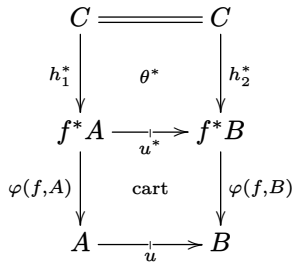
be a tabulator over $1^{\bullet}_{M}$ and $1^{\bullet}_{f}\colon 1^{\bullet}_{N} \longrightarrow 1^{\bullet}_{M}$ be a cell in $\mathbb{B}$. Consider a double cone over π_u given by



with $k \circ c_T = h_2$ and $l \circ c_T = h_1$. Clearly, the cell

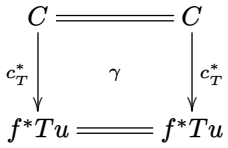


lies over $1^{\bullet}_{f}$ in $\mathbb{B}$. Let

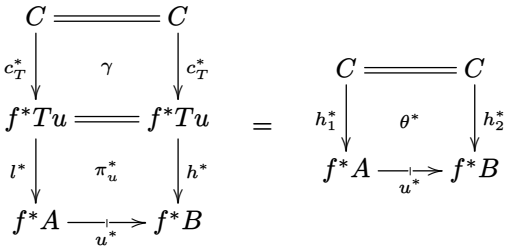


be the vertical-cartesian factorization of θ. Since every fibre has a tabulator, there is a

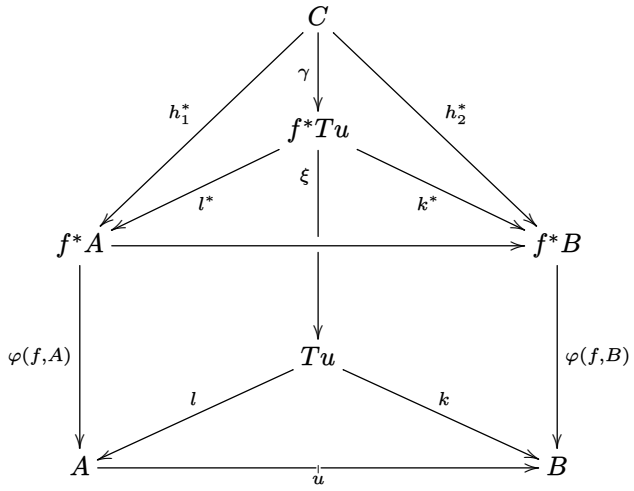
vertical cell



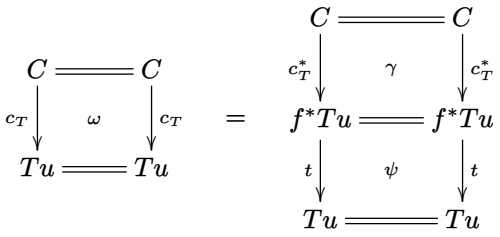
such that



We obtain the following diagram



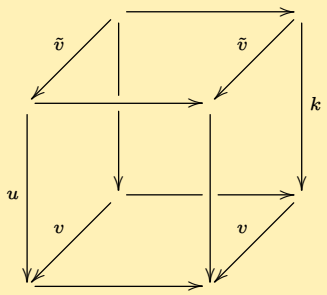
The cell



is the desired mediating cell. □

I have made this interesting observation:

Lemma 11.9. For a a strict double fibration $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$ of double categories with tabulators. For a commuting cube



given by $(\mathbf{T}), (\mathbf{F}), (\mathbf{B}), (\mathbf{L}), (\mathbf{R}), (\mathbf{G})$ as the top, front, back, left, right, and bottom faces respectively, we have the following

1. If $(\mathbf{B})$ is tabulator cell and $(\mathbf{G})$ is cartesian, then $(\mathbf{T})$ is cartesian
2. If $(\mathbf{F})$ is a tabulator and $(\mathbf{G})$ is cartesian then $(\mathbf{B})$ is a tabulator if and only if $(\mathbf{T})$ is cartesian.
3. If $(\mathbf{T})$ and $(\mathbf{B})$ are tabulator cells and $(\mathbf{G})$ is cartesian, then the cube is a pullback.

Remark 11.10. Condition 2 in Lemma 11.9 is a Beck-Chevalley like condition for fibred tabulators.

Remark 11.11. In the above formulation, we require the source s and target t maps to preserve the fibre structure. Moreover, each property of a double fibration (i.e., terminal objects or pullbacks) should lie in the 2-category of fibrations with terminal objects or pullbacks. Since tabulators only appear in the double categorical case, the treatment is purely double categorical.

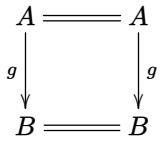
Since we gave a characterization theorem for double terminals, we also have the following characterization theorems for pullbacks and tabulators.

Theorem 11.12. A double fibration $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$ is a double fibration of double categories with pullbacks if and only if $\mathbb{E}$ has and $\mathcal{P}$ preserve them.

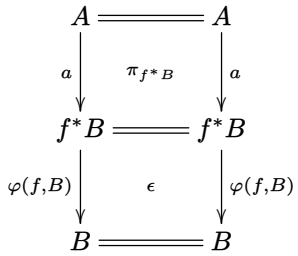
This follows readily from the double fibration structure since it holds for $\mathcal{P}_0$. However, the following theorem does not.

Theorem 11.13. Let $\mathbb{B}$ have tabulators. A double fibration $\mathcal{P}\colon \mathbb{E} \longrightarrow \mathbb{B}$ is a double fibration of double categories with tabulators if and only if $\mathbb{E}$ has tabulators and $\mathcal{P}$ preserves them.

Proof. Let



be a cell in $\mathbb{E}$. Consider the following diagram



where ϵ is cartesian over $1^\bullet_f \colon 1^\bullet_N \longrightarrow 1^\bullet_M$ and π_{f^*B} is a tabulator in the fibre. Then π_{f^*B} is a tabulator in $\mathbb{E}$ by Lemma 11.8 and ϵ is a tabulator in $\mathbb{E}$ by Lemma 11.7. It remains to see that the composite $\frac{\pi_{f^*B}}{\epsilon}$ is a tabulator in $\mathbb{E}$.

□

12 Dual Notions

12.1 2-Opfibrations

Definition 12.1. A 2-functor $P \colon \mathcal{E} \longrightarrow \mathcal{B}$ is a 2-opfibration if the following hold

1. For any $e \in \mathcal{E}$ and $u \colon Pe \longrightarrow b$, there is an opcartesian lift $\bar{u} \colon e \longrightarrow u_! e$ with $P\bar{u} = u$;
2. for any $g \in \mathcal{E}$ and $\alpha \colon Pg \Longrightarrow h$, there is an opcartesian lift $\bar{\alpha} \colon g \Longrightarrow \alpha_! g$ with $P\bar{\alpha} = \alpha$; and
3. Horizontal composition of opcartesian 2-cells is again opcartesian.

Remark 12.1. The above definition refers to coop-2-fibrations as 2-opfibrations. Comparing to Hermida, opcartesian 1-cells and 2-cells would be defined using pre-composition instead of post-composition.

12.2 Double Bifibrations

A strict double functor $\mathscr{P} \colon \mathbb{E} \longrightarrow \mathbb{B}$ defines a double fibration if the components P_0 and P_1 are fibrations. We say $\mathscr{P}$ is a double fibration if in addition the components P_0 and P_1 are opfibrations.

12.2.1 Relation to 2-fibrations

Proposition 12.2. A 2-functor $\mathscr{P}\colon \mathcal{E} \longrightarrow \mathcal{B}$ is a 2-fibration if and only if $Q(\mathscr{P})\colon Q(\mathcal{E}) \longrightarrow Q(\mathcal{B})$ is a double fibration.

Proof. ($\Longrightarrow$) The cartesian 1-cells make the functor $Q(P_0) = |P|\colon |\mathcal{E}| \longrightarrow |\mathcal{B}|$ an ordinary fibration. We are required to show: $Q(P)_1$ is a fibration and the external source and target maps of $Q(\mathcal{E})$ are cleavage-preserving. To this end suppose there is an object $X \nrightarrow Y$ in $Q(\mathcal{E})_1$ and a morphism

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ {\scriptstyle f}\downarrow & \Downarrow\alpha & \downarrow{\scriptstyle g} \\ PX & \xrightarrow[Pv]{} & PY \end{array}$$

in $Q(\mathcal{B})_1$. The lift of α in $Q(\mathcal{E})_1$ can be constructed as follows: Since $\mathscr{P}$ is locally a fibration, the cartesian lift $\tilde{\alpha}$ of α exists. Since $\sigma(g, Y)$ is cartesian, there is a unique $\tilde{u}\colon f^*X \longrightarrow g^*Y$ such that the following triangle commutes

$$\begin{array}{ccc} f^*X & \overset{\tilde{u}}{\nrightarrow} & g^*Y \\ {\scriptstyle \sigma(f,X)}\downarrow & \Downarrow\tilde{\alpha} & \downarrow{\scriptstyle \sigma(g,Y)} \\ X & \xrightarrow[v]{} & Y \end{array} \qquad (\clubsuit)$$

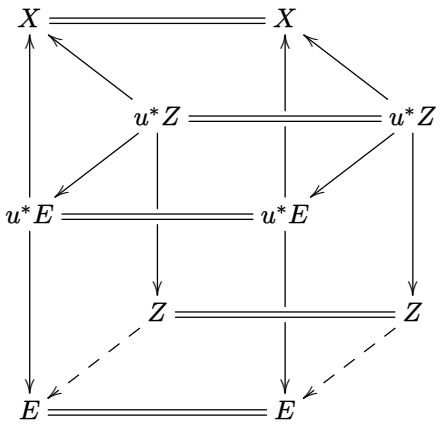
Claim: ($\clubsuit$) is cartesian w.r.t $Q(P)_1$

Proof. Since the vertical arrows are cartesian and the 2-cell is cartesian, this means ($\clubsuit$) will be $Q(P)_1-$cartesian. □

□

13 Double Internal Products

Double Evaluation Spans



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